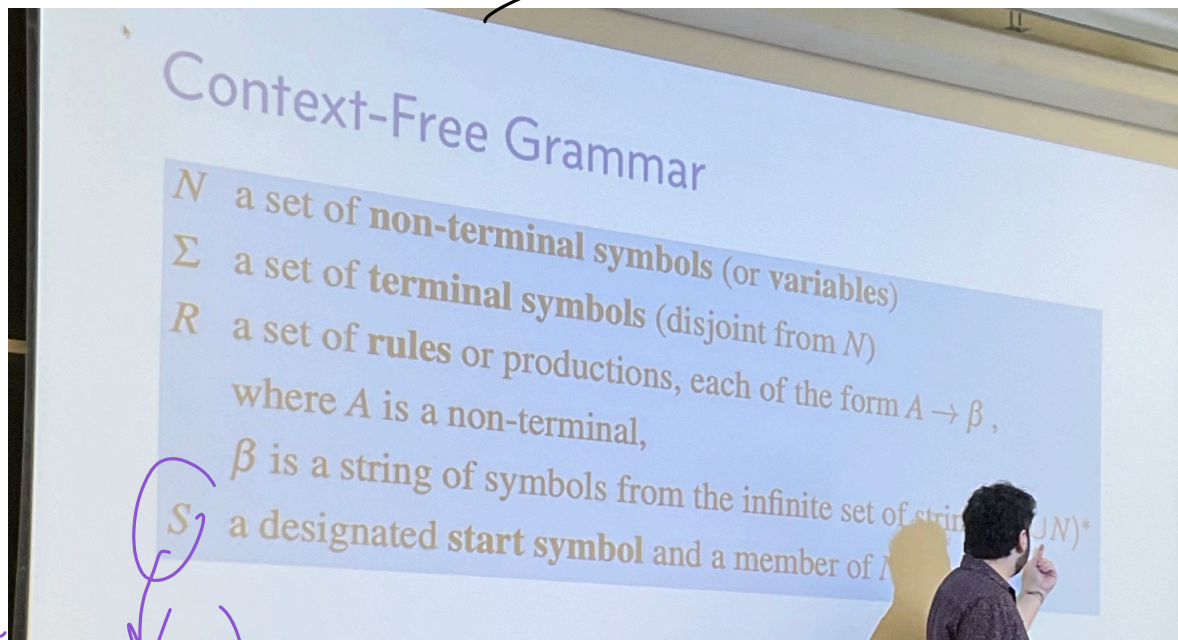
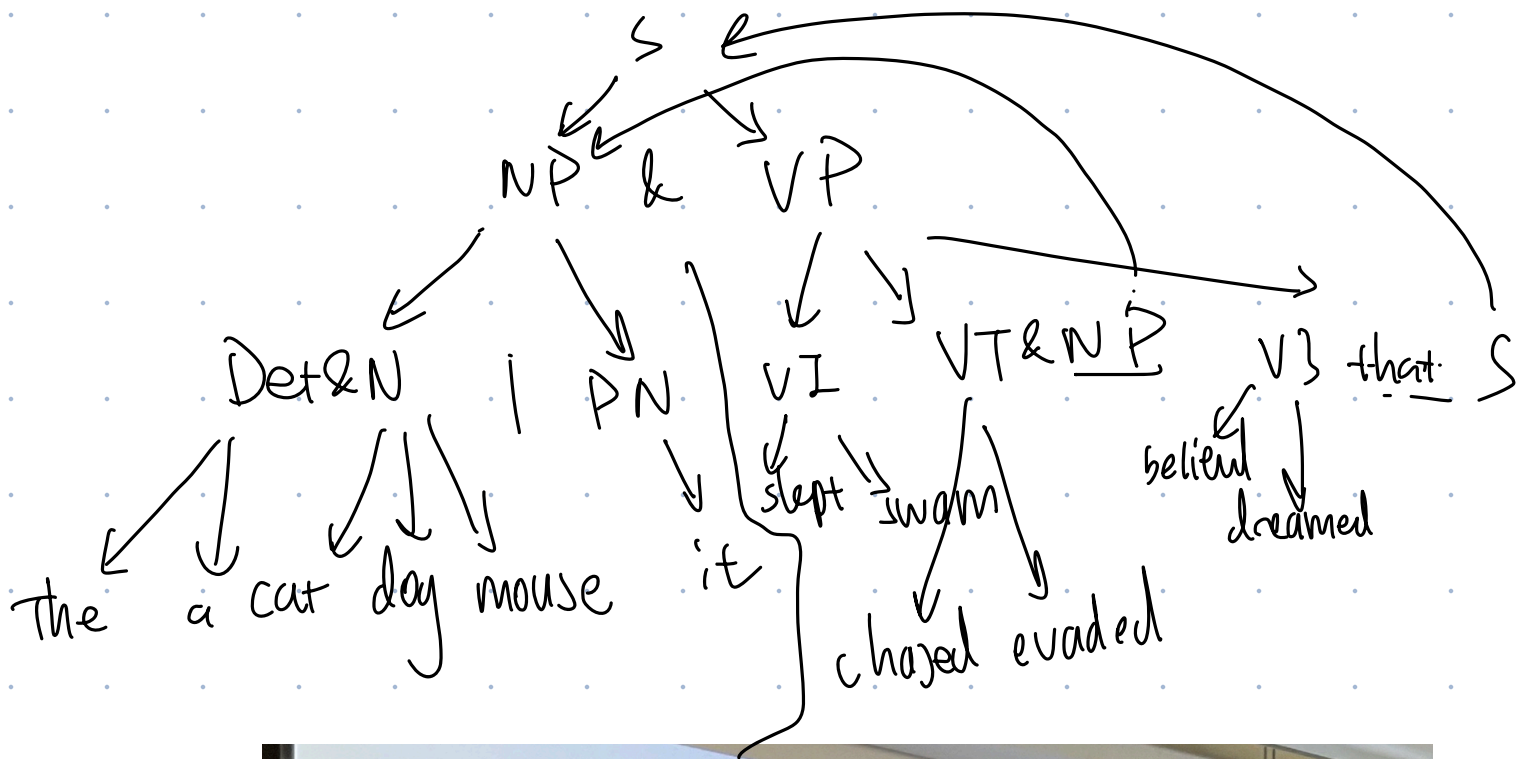




S $\rightarrow$ **NP VP**
NP $\rightarrow$ **Det N | PN**
Det $\rightarrow$ **the | a**
N $\rightarrow$ **cat | dog | mouse**
PN $\rightarrow$ **it**
VP $\rightarrow$ **VI | VT NP | V3 that S**
VI $\rightarrow$ **slept | swam**
VT $\rightarrow$ **chased | evaded**
V3 $\rightarrow$ **believed | dreamed**

Context-Free Grammar

if $A \rightarrow \beta$ is a production of R and α and γ are any strings in the set $(\Sigma \cup N)^*$, then we say that $\alpha A \gamma$ **directly derives** $\alpha \beta \gamma$, or $\alpha A \gamma \Rightarrow \alpha \beta \gamma$.

$S \rightarrow$	$NP VP$
$NP \rightarrow$	$Det N \mid PN$
$Det \rightarrow$	$the \mid a$
$N \rightarrow$	$cat \mid dog \mid mouse$
$PN \rightarrow$	it
$VP \rightarrow$	$VI \mid VT NP \mid V3 \text{ that } S$
$VI \rightarrow$	$slept \mid swam$
$VT \rightarrow$	$chased \mid evaded$
$V3 \rightarrow$	$believed \mid dreamed$

the cat VP $\Rightarrow$ the cat V3 that S

*the cat V3 that S $\Rightarrow$
the cat dreamed that S*

Context-Free Grammar

Derivation is then a generalization of direct derivation:

Let $\alpha_1, \alpha_2, \dots, \alpha_m$ be strings in $(\Sigma \cup N)^*$, $m \geq 1$, such that

$$\alpha_1 \Rightarrow \alpha_2, \alpha_2 \Rightarrow \alpha_3, \dots, \alpha_{m-1} \Rightarrow \alpha_m$$

We say that α_1 **derives** α_m , or $\alpha_1 \Rightarrow^* \alpha_m$.

$S \rightarrow$	$NP VP$
$NP \rightarrow$	$Det N \mid PN$
$Det \rightarrow$	$the \mid a$
$N \rightarrow$	$cat \mid dog \mid mouse$
$PN \rightarrow$	it
$VP \rightarrow$	$VI \mid VT NP \mid V3 \text{ that } S$
$VI \rightarrow$	$slept \mid swam$
$VT \rightarrow$	$chased \mid evaded$
$V3 \rightarrow$	$believed \mid dreamed$

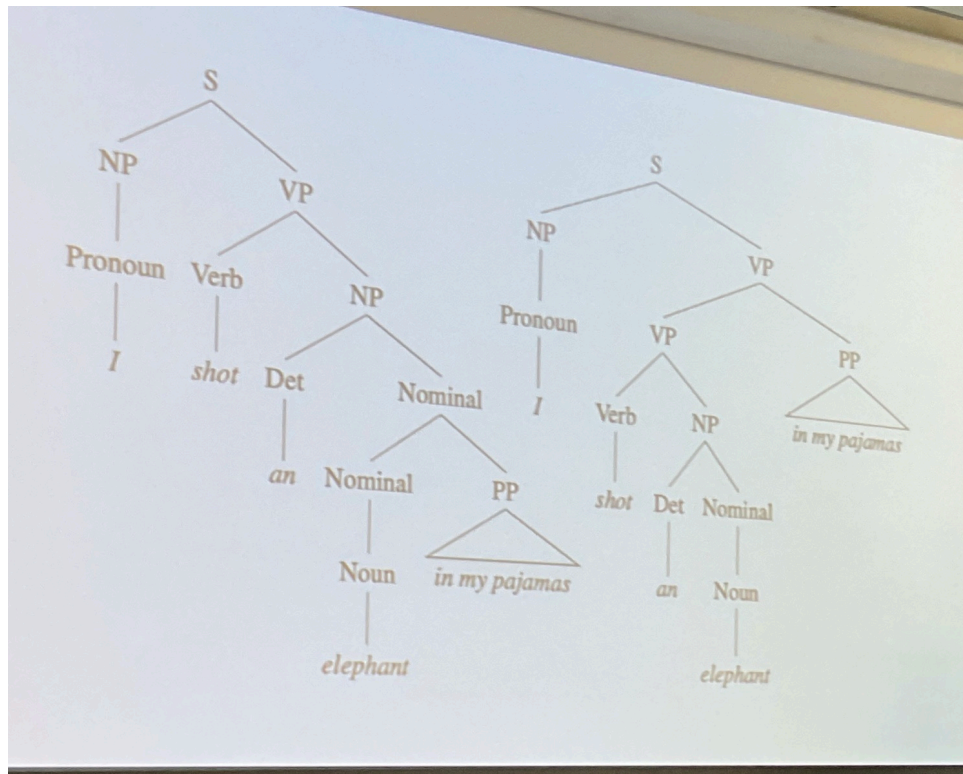
the cat VP $\Rightarrow^$ the cat dreamed that S*

S $\Rightarrow^$ the dog evaded the cat*

Pushdown automata can determine the validity of context-free grammar.

Parse trees -

↳ Multiple trees $\Rightarrow$ ambiguity.



I didn't understand the assembly language parse tree thing. Have to think abt it more.

Genre Random String in HTML

↳ can be creative

↳ course titles / new titles

Friday Review sessions:
Friday 3:10 PM A.M.